

RESTAURANT SALES ANALYSIS

A Project Report

*Submitted to the APJ Abdul Kalam Technological University
in partial fulfillment of requirements for the award of degree*

Bachelor of Technology

in

Computer Science and Engineering

by

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May 2024

DECLARATION

We hereby declare that the project report **RESTAURANT SALES ANALYSIS**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Prof. Jini Raju

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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Abstract

This project focuses on Exploratory Data Analysis of sales dynamics through advanced statistical methods and trend analysis models. Beginning with meticulous data preprocessing, including data quality checks and handling missing values, the project ensures a robust foundation for subsequent analyses. The project further extends its reach to evaluating the success of high-performing items, addressing underperformers, and examining the impact of product lifecycle changes. External factors like weather conditions, social events, and marketing campaigns will be systematically analyzed to recognize their influence on sales dynamics. Sales analysis will be leveraged for accurate sales forecasting and optimized inventory management. Visualization tools will be implemented to facilitate accessible data interpretation, empowering stakeholders to make informed decisions and adapt strategies for sustained success. This multifaceted approach, encompassing data preprocessing, statistical methods, and trend analysis, aims to provide a nuanced understanding of sales dynamics and enable strategic decision-making for long-term business success.

Keywords: Exploratory Data Analysis, Trend Analysis, Seasonality

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Chapter 1

Introduction

1.1 General Introduction

The restaurant industry has been one of the major contributors to food wastage in India. Previous studies have shown that 25-30% of the food prepared by the restaurants in India tend to get wasted. The restaurant industry faces a loss of staggering amount of INR 5000 crore. According to the Food Waste Index Report 2021 by the United Nations Environment Programme (UNEP), India generates about 68.7 million tons of food waste per year, out of which 11.9 million tons are generated by the food service sector. The main reasons to this include lack of storage facilities and transportation infrastructure. During COVID pandemic, food sector was the worst hit sector in India, and even globally. Food wastage increased by another 25-30% since the supply chains were disrupted. Food service industry went through a massive decline of 53% in 2021 compared to the previous year. This has led to the permanent closure of over 25% of restaurants all over India resulting in loss of jobs of almost 24 Lakh people, as per the report published by National Restaurant Association of India (NRAI). Other crucial contributors of food wastage in restaurant industry include overproduction, large portion sizes of food items served, menu variety and customer satisfaction. Figuring out the items that are least popular among the customers, i.e. the items that are least sold or not sold at all, can be a way to control excessive food wastage in restaurants thus boosting its sales. Restaurant sales are important to the respective restaurants to ensure that they are doing well in their business. They are mainly based on the revenue generated by the restaurants with respect to every item sold. Thus, customer satisfaction is also a crucial factor that decides the success of the restaurant. Higher sales of a certain item proves that it is more popular among the customers compared to the rest of

the items. However, food wastage has been an inevitable factor that contributes to the downgrade of the flourishing business of various restaurants. Due to excessive food wastage, it affects the revenue of the restaurant.

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External factors such as weather conditions, social events, public holidays and marketing campaign, also affect the restaurant sales. So, it is equally important to take these factors into consideration while performing the data analysis. Examining the shifts and fluctuations in the production of items in the restaurant can help the stakeholders in managing financial resources, production of food resources, managing the required raw materials and inventory levels with optimal efficiency. This project thus mainly focuses on using raw sales data to assess the trends and sales shifts throughout a time period, in order to help the restaurant managers flourish in their business, reducing the recurring food wastage. Thus, an accurate and detailed data analysis for products can be done to achieve this. This project has used Exploratory Data Analysis (EDA) for this purpose.

Exploratory Data Analysis or EDA is a crucial step in the data analysis process. It involves the initial investigation of data to discover patterns, trends, anomalies, and relationships. The primary goal of EDA is to gain insights into the structure of the data and to formulate hypotheses that can guide further analysis. It can calculate and examine descriptive like mean, median and mode to help the user understand better about the data used. It can also create visual representations of the data in the form of graphs and can even assist the user in figuring out and adding in the missing values to the data. EDA can find out the anomalies in the data as well. It can investigate further into the relationship between the variables and attributes and can search for pattern investigation using various statistical and machine learning techniques. EDA is an iterative process that helps data analysts and scientists understand their data before applying more complex modelling techniques. Thus, EDA has been focused upon in this project since it encompasses a variety of statistical, visual, and exploratory techniques, ensuring a comprehensive understanding of the dataset before moving

on to more advanced modelling and analysis. Therefore, it aims to distinguish high performing items and the underperformers, providing insights crucial for product-level decision-making.

1.2 Motivation

Opening and running a restaurant isn't an easy job. Thin margins and evolving and oscillating food trends make it difficult. However, competition from a variety of other dining options is a very difficult aspect in running a restaurant. In fact, 1 out of 5 restaurants named tough competition a top challenge in 2018. Fortunately, there are ways to be successful despite this raging situation. 85% of the customers don't know what to order hours beforehand. In India, according to a survey conducted in 2021, 24% of Indians were vegetarians and 18% were selective meat eaters. Similarly, 9% were vegans and around 8% were pescatarians (doesn't eat meat except for fish). Due to this diversity in customer likings and the budding competition in the food sector, it is important to conduct a data-driven decision making with context to restaurant sales analysis. As the restaurant industry continually evolves, the need to extract meaningful insights from sales data becomes necessary. Hence, this project addresses the situation by making use of sophisticated data analysis techniques like Exploratory Data Analysis (EDA) to reveal the hidden patterns and trends within restaurant sales. This can even be helpful in making strategic decisions, ranging from optimizing menu options and their prices and identifying high performing or popular dishes to adapting to external factors like events, weather conditions, social events, promotional campaigns etc. By figuring out and understanding these insights and the impact of product changes on sales, the project ensures to empower stakeholders with actionable intelligence, guiding them towards more informed, adaptive, and efficient management practices. Ultimately, it enhances the competitiveness and sustainability of restaurants in a dynamic market through the transformative power of data analysis

1.3 Problem statement

To use data analysis techniques for comprehensive insights into sales dynamics, focusing on the identification of trends, seasonality, and the performance of individual and cluster of items. Through statistical methods and trend analysis, uncover patterns, detect peak seasons, and understand dayspecific variations. The analysis extends to evaluating the success of high-

performing items and addressing underperformers, while also assessing the impact of newly added or discontinued products. External factors such as weather conditions, social events, and marketing campaigns will be examined to recognize their influence on sales. Furthermore, predictive analytics will be used for accurate sales forecasting and optimized inventory management. Implement visualization tools for accessible data interpretation, empowering stakeholders to make informed decisions and adapt strategies for sustained success.

1.4 Objectives

- To conduct Exploratory Data Analysis (EDA) on the restaurant sales data to find the highs and lows of different items provided by the restaurant
- To analyse the dataset to identify trends, patterns, and seasonality in sales data of the restaurant
- To use statistical methodologies to gain insights into the variability and fluctuations in the sales metrics
- To figure out how different types of external factors such as weather conditions, social events, public holidays, marketing campaigns etc. affect the sales of the respective restaurant
- To visualize the data insights and ensure that the stakeholders can easily interpret and extract meaningful information about the dataset acquired
- To evaluate the performance of individual items through historical sales data and identify the high-performing items and the least popular ones

1.5 Chapter Organization

The remainder of this report is laid out in the following manner. In Chapter 2, a detailed examination of the literature works on this subject is explained. The methodology of the models used is specified in Chapter 3, and the task division and allocation are discussed in Chapter 4. Chapter 5 concludes with some thoughts on the work's future potential. The references used for the research are listed in Chapter 6, providing a comprehensive bibliography of the sources consulted and cited throughout the research report.

Chapter 2

Literature Survey

2.1 Related Works

The study [1] specifies the critical role of accurate demand forecasting in managing restaurants and staff canteens. The main focus is on predicting future sales of menu items to optimize food stock ordering, minimize pre-consumer food waste, and ensure restaurant profitability. The proposed forecasting approach relies solely on Point of Sales (POS) system data, featuring two generalized additive models for predicting future sales. The evaluation, conducted on data from a casual restaurant and a large staff canteen with multiple time series spanning 20 months, addresses challenges such as strong seasonalities, trend changes, data gaps, and outliers. The paper highlights the environmental and economic impact of food waste in the industry, stressing the importance of reducing waste to enhance profitability and mitigate environmental concerns. The authors assume independence of sales for individual menu items and differentiate between the absence of sales due to customers' lack of interest and various other reasons. The research advocates for a data-driven approach tailored to the restaurant industry's short lead times and simple supply chains, emphasizing the significance of utilizing POS data for forecasting.

The paper [2] proposes a demand forecasting model for restaurants using machine learning, aiming to enhance the accuracy of predictions by considering various factors such as store location, weather, and events. While previous research has focused on demand forecasting using Point of Sale (POS) data, the authors emphasize the need for store-specific models that account for multiple variables. The laborintensive nature of the service industry, particularly in restaurants, poses challenges in improving labor productivity. The paper discusses the construction of a

demand forecasting model that integrates data from POS, store location, weather, and events using machine learning techniques. The goal is to advance store management by optimizing employee work arrangements and food material ordering based on accurate customer forecasts. The authors explore the combination of internal data (POS) and external data to enhance the precision of demand forecasting. The paper outlines the approach, methodology, and verification results of the machine learning-based demand forecasting model using real store data. The research contributes to addressing challenges in face-to-face service industries like restaurants and emphasizes the importance of considering various factors in demand forecasting for improved store management.

The paper [3] looks into the significance of food sales prediction for optimizing profits and minimizing food wastage in companies. Two key objectives are outlined: comparing datasets with high and low feature correlation and employing various machine learning algorithms to identify the top three performers. The study distinguishes itself by simultaneously using multiple datasets and diverse algorithms, emphasizing the role of artificial intelligence in handling large datasets. Preprocessing steps are applied to the datasets, and ten machine learning algorithms are evaluated based on metrics such as root mean square error and mean square error. The study even highlights the importance of accurate sales prediction for reducing losses and enhancing profits, providing specific algorithm recommendations for datasets with different correlation levels.

The paper [4] focuses on weather prediction through data mining techniques applied to meteorological data collected between 2010 and 2014 in Chennai, India. Utilizing the CRISP DM framework, the study involves data understanding, cleanup, and transformation. Descriptive and exploratory data analysis is conducted to identify daily maximum and minimum temperatures and assess correlations between weather attributes. Clustering techniques are employed to classify weather attributes, and time series analysis is performed using neural network models to forecast future temperatures. The evaluation of the model, based on different metrics, provides insights into the effectiveness of the predictive approach. The paper underscores the significance of accurate weather prediction for applications such as disaster management, agriculture, and construction.

The paper [5] presents a case study employing various machine learning (ML) models, including recurrent neural network (RNN) models, for accurate sales forecasting in a mid-sized restaurant using real-world sales data. Three different datasets are generated to assess the impact of trend and seasonality on the forecasting models. The study emphasizes feature engineering and optimal feature subset selection to enhance forecasting. Evaluation of models is conducted for one-day and one-week forecasting periods. Linear models and RNN models such as LSTM

and TFT perform well in one-day forecasting, while non-RNN models underperform in one-week forecasting compared to RNN models. It also explains the importance of accurate sales predictions for employee scheduling in small to medium-sized restaurants and explores various forecasting methods, highlighting the limitations of traditional approaches. The study contributes to the understanding of the effectiveness of modern ML techniques in restaurant sales forecasting, comparing them with traditional statistical methods like ARIMA. The results provide insights into the landscape of forecasting methodologies, emphasizing the need for accurate predictions for efficient restaurant management.

The paper [6] focuses on the crucial aspect of demand forecasting for successful restaurant yield and revenue management systems. It provides a comprehensive literature review and classification of various techniques employed in restaurant sales and consumer demand forecasting over the past two decades. The analysis categorizes these techniques into seven distinct categories, including hybrid models, and briefly describes the methodologies, advantages, drawbacks, and relevant research papers associated with each category. The significance of sales forecasting is emphasized for both independent restaurants and restaurant chains, highlighting its impact on operational and product management at the store level and its contributions to corporate-level strategic planning. The historical reliance on judgmental forecasting methods in the restaurant industry is acknowledged, and the paper explores a diverse array of models designed to enhance forecasting accuracy by considering factors such as time, weather conditions, economic influences, and random cases.

2.2 Limitations

The study [1] primarily focuses on the restaurant industry and may not generalize well to other industries with different characteristics. Additionally, the evaluation relies on cross-validation, which might not capture all possible scenarios and variations in real-world applications. The choice of evaluation metrics may also influence the interpretation of model performance, and the study acknowledges the importance of considering both scale-dependent and scale-invariant measures. Finally, the performance comparison is based on a specific set of forecasting models, and there may be other models or techniques that were not considered in the evaluation.

In study [2], The authors express future plans to enhance forecasting accuracy and delve into researching the efficiency of store management, including automated food material ordering and

work arrangement based on forecasting results.

In the paper [4] the analysis focuses on weather attributes measured at 2m above the ground, potentially overlooking variations at different altitudes. A limitation arises from the exclusive reliance on the CRISP Data Mining framework, and alternative frameworks or methodologies could offer diverse perspectives. The clustering technique, specifically K-means, may be sensitive to the initial choice of centroids and may not be suitable for all types of data distributions. Additionally, the forecasting model's performance is evaluated based on specific criteria, and other relevant metrics or comparisons with alternative forecasting models could enhance the comprehensiveness of the assessment. Future research directions may involve exploring apparent weather experiences at varying altitudes above the ground and employing different forecasting models to improve overall accuracy.

The research [5] encounters limitations in achieving a scalable model for prediction horizons of at least one week using real-world data. While recurrent neural network (RNN) models, particularly the Temporal Fusion Transformer (TFT) model, demonstrated promising performance, challenges persisted in achieving consistent results in both one-day and one-week forecasting. The difficulty in scaling traditional models like kernel ridge algorithms to one-week forecasting suggests the need for further investigation into stepwise forecasting approaches. Transformer-based models, including TFT, show potential for improvement, but the study acknowledges the narrow scope of data from a single restaurant location. The need for experimentation with diverse and larger datasets, including considerations like weather data, is highlighted for future research to enhance feature engineering and forecasting accuracy.

Chapter 3

Proposed Methodology

3.1 Formulation & Presentation of Problem

Examining the delicate details of sales data of restaurants is an activity that should be started up on initially. Its main objective is to find insightful information that can guide the stakeholders into making better strategic choices and decisions. This involves the use of Exploratory Data Analysis or EDA in order to look into the sales dynamics. This focuses on identifying trends and seasonalities and to comprehend the performance of individual and cluster of items. Restaurants face a lot of challenges mainly consisting of ineffective inventory management, which can lead to food waste from overstocking and overproduction. Hence, it is important to uncover patterns, detect peak seasons and understand the variations on a daily basis. This can help the restaurant management sector to make data-driven decisions thus contributing to controlling overproduction and food wastage. From the data analysis, the success of high performers can be evaluated and the issue of underperformers can be addressed as well. This will assist the stakeholders in deciding to produce the popular items on a larger rate compared to the rest. External factors like weather conditions, social events, public holidays and marketing campaigns also play a major role in influencing the sales of a restaurant. Hence these factors are taken into consideration for the same.

However, several obstacles have been recognized when dealing with this problem:

- **Multifaceted Temporal Dynamics:** Restaurants face the challenge of managing temporary patterns influenced by factors such as time of the day, day of the week and seasonal variations. Therefore, EDA takes these little details into consideration when performing the data analysis. The analysis also takes into account any irregularities, anomalies or unexpected

patterns within the seasonality. This provides a better output and can be helpful in demand forecasting of restaurant sales data, which will be done in the future. This can also help the management in producing the food accordingly, thus, contributing largely into reduction of food wastage.

- **Varying sales outcomes for different items:** Usually, the sales of different items can be different. This means that when the sales of one item is high at one point, the sales of another can be pretty low, or vice-versa. This can pose a problem during data analysis since such a delicate issue can have an adverse effect in producing an item on larger scale. Thus, effective evaluation of performance of individual items is necessary, since it impacts both inventory management and strategic planning.
- **Impact of external factors:** External factors such as weather conditions, social events, public holidays and marketing campaigns can have a large influence on the sales of a restaurant. There can even be days when the restaurant isn't open. All these factors should be taken into consideration when performing EDA, since these factors contribute to sales fluctuations and can complicate precise measurement.
- **Product Impact:** There can be a change in the sales pattern whenever a new product is introduced and whenever a product is discarded or removed from the menu. This can have a major effect on the total sales of the restaurant and the customer satisfaction rate. Hence, it is important to look into the customer feedback and figure out the most popular items and the least popular ones through thorough data analysis.
- **Cluster Dependency Oversight:** It is true that an item can individually influence the restaurant sales data. However, the fact that a category of food items as a whole can have an effect on its sales as well. There can be different categories of food items such as veg, non-veg, beverages, vegan etc. The performance of each category can vary from restaurant to restaurant. Customers are usually tended to go for the category that the restaurant is popularly known for. In short, customer preferences and ordering behaviours are often shaped by broad cluster compared to singular items. Hence, performing an in-depth analysis of restaurant sales and applying the findings to demand forecasting or prediction is a powerful strategy that can help address this situation.
- **Temporal Analysis:** : It is important to look into the patterns and trends that occur in a

restaurant over time. This can help the stakeholders in deciding production of which all items can contribute largely into the success of its business and reduction of food wastage. Time series analysis can be used for the same, helping the user to uncover recurring patterns, over different time intervals.

- **Item Performance Evaluation:** Identifying high performing items and low performing items of a restaurant is essential for the management to run the restaurant efficiently. This way, the restaurant can produce the popular dishes on a larger scale and reduce the production of the less popular ones or maybe even cease their production provided that their production is really low and that they don't contribute much into the success of the restaurant business. Hence, figuring out the most popular and the least popular items can enable the businesses to allocate resources efficiently, focus promotional efforts on high performing items and discontinue the underperformers.
- **Visualization:** The data analysis of the restaurant sales should let the user visualize the data provided. Graphs such as pie charts, bar graphs etc. can be produced as an output, which can help the stakeholders to interpret and get an insightful information about the dataset used.
- **Accurate demand prediction:** After the analysis of the data, accurate prediction of future sales of the restaurant can be performed. Time series models can be used for sales forecasting. This can be done by incorporating all the external factors and by taking all the above challenges and insights into consideration.

3.2 Solution Approach

The solution approach encompasses a systematic methodology tailored to comprehensively analyze sales dynamics, starting with rigorous data preprocessing to ensure data integrity and quality. Utilizing advanced statistical methods and trend analysis models, this approach delves into identifying intricate trends, seasonality patterns, and day-specific variations within the sales data through thorough Exploratory Data Analysis (EDA). The focus extends to evaluating the performance of individual items, discerning high-performing products, addressing underperformers, and assessing the impact of product lifecycle changes. External factors like weather conditions, social events, and marketing campaigns undergo meticulous analysis to uncover their influence on sales dynamics. Leveraging these insights, the approach integrates predictive analytics for

accurate sales forecasting and optimized inventory management. Implementing visualization tools aids in presenting accessible and meaningful data interpretations, empowering stakeholders to make informed decisions and adapt strategies for sustained success. This multifaceted approach, encompassing EDA, statistical methods, trend analysis, and predictive modeling, aims to provide nuanced insights into sales dynamics and facilitate strategic decision-making for long-term business success.

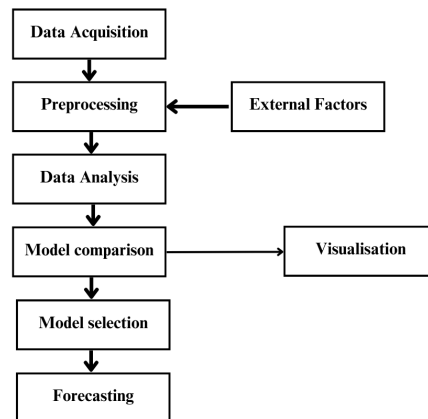


Figure 3.1: Flow Diagram

3.2.1 Data Processing and Quality Checks

- **Data Cleansing and Imputation:** Perform meticulous data cleaning to address missing values, outliers, and inconsistencies within the sales dataset, ensuring data integrity for subsequent analyses.
- **Normalization and Standardization:** Normalize and standardize data if necessary to maintain consistency across different features, facilitating accurate comparisons and analyses.
- **Feature Engineering:** Create new features or transform existing ones to enhance the dataset, incorporating relevant temporal features, categorical encodings, or derived metrics crucial for trend analysis and predictive modeling.

3.2.2 Exploratory Data Analysis (EDA) for Sales Dynamics

- **Trend Identification:** Utilize statistical methods and visualization techniques to identify long-term trends, seasonality, and day-specific variations within the sales data, providing insights into sales dynamics.
- **Pattern Recognition:** Employ EDA to uncover patterns and correlations, enabling a deeper understanding of sales fluctuations, peak seasons, and recurring sales behaviors.
- **Statistical Analysis:** Conduct statistical analyses to explore the distribution and variability of sales metrics, identifying outliers or anomalies that could impact subsequent analyses.

3.2.3 Item Performance Analysis and Cluster Dynamics

- **Performance Evaluation:** Analyze historical sales data to evaluate individual item performance, identifying high-performing items and underperformers. Understand the contributing factors behind their success or challenges.
- **Cluster Analysis:** Group items based on shared characteristics and analyze sales dynamics within each cluster, recognizing common sales behaviors or preferences that might influence strategic decisions.

3.2.4 Impact Assessment of External Factors

- **External Factor Analysis:** Investigate the influence of external factors like weather conditions, social events, and marketing campaigns on sales patterns. Establish correlations between these factors and variations in sales performance using statistical methods or correlation analysis.
- **Causal Inference:** Determine the causal impact of external factors on sales dynamics, distinguishing correlations from actual causal relationships, if applicable.

3.2.5 Evaluation of Product Changes

- **Product Lifecycle Impact:** Assess the impact of newly introduced or discontinued products

on overall sales and individual item performance. Identify shifts in customer behavior or preferences following these changes.

3.2.6 Visualization Tools Implementation

- **Interactive Dashboards:** Develop user-friendly and interactive visualization tools (dashboards, graphs, charts) to present EDA findings, sales trends, item performance, and the impact of external factors. These tools enable stakeholders to extract actionable insights easily.

3.2.7 Predictive Analytics for Sales Forecasting

- **Time Series Forecasting:** Employ predictive models (Holt-Winters, time series models) incorporating EDA insights and external factors to accurately forecast sales, aiding in inventory management and strategic planning.

3.2.8 Iterative Improvement and Adaptation

- **Model Refinement:** Continuously refine models based on new data or changing trends to enhance forecasting accuracy.
- **Feedback Integration:** Incorporate feedback from stakeholders based on data interpretations to adapt strategies and decision-making processes for sustained success.

3.3 Software/Hardware Requirements and Specifications

Software:

- Programming Language: Python (libraries like Pandas, NumPy, Scikit-learn)
- Programming language : Python(with libraries like Scikit-learn,pandas,NumPy)
- Data Visualisation: MatPlot library in python
- Database : in google excel or google csv format

- Dataset : should contain Date of sale, time of sale, Item sold, quantity of item sold

Hardware:

- Processor : Multicore CPU for training model
- Memory : 4GB or more
- Storage : preferred SSD storage for faster access

3.4 Implementation (Design & Coding)

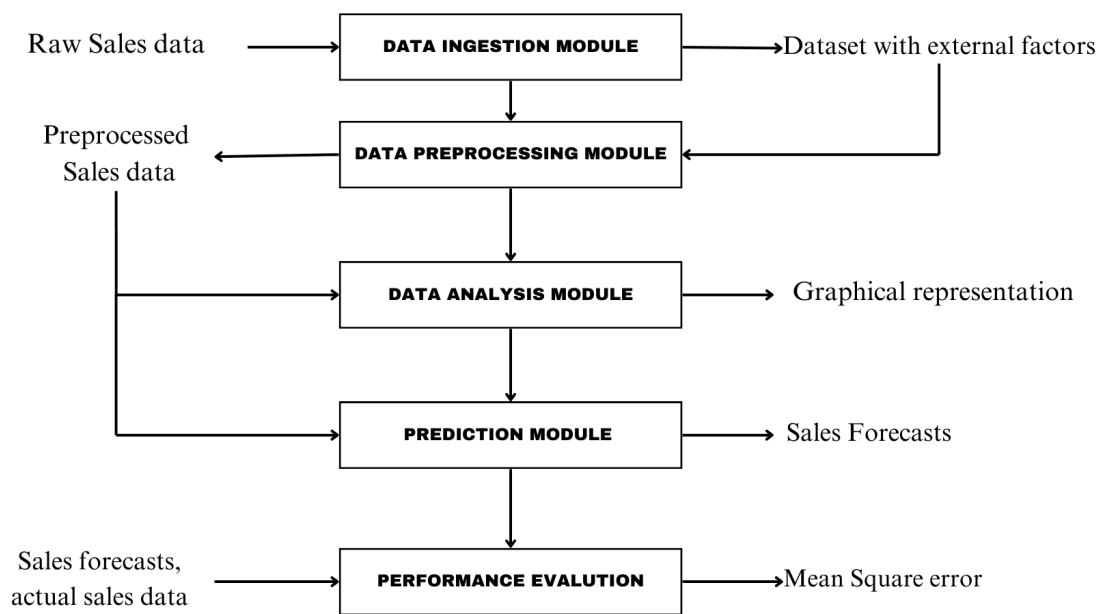


Figure 3.2: Architecture Diagram

3.4.1 Data Acquisition

- The Data acquisition stage is where the datasets of restaurants from different sources are acquired which would be used for prediction and analysis. A lot of restaurants were requested to lend us the dataset for the same. Unfortunately, they were listed as confidential, hence the team couldn't acquire them. Thus, online platform was considered an alternative for data acquisition. Since synth data didn't work efficiently as required, a dataset from Kaggle was chosen to implement this.

date	Bill Numbe	Item Desc	Quantity	Rate	Tax	Discount	Total	Category
04-01-2010 13:15	G0470115	QUA MIN	1	50	11.88	0	61.88	BEVERAGE
04-01-2010 13:15	G0470115	MONSOOF	1	100	23.75	0	123.75	BEVERAGE
04-01-2010 13:17	G0470116	MASALA C	1	40	9.5	0	49.5	BEVERAGE
04-01-2010 13:19	G0470117	QUA MIN	1	50	11.88	0	61.88	BEVERAGE
04-01-2010 01:20	G0470283	MOROCCA	1	45	10.69	0	55.69	BEVERAGE
04-01-2010 01:20	G0470283	QUA MIN	1	50	11.88	0	61.88	BEVERAGE
04-01-2010 13:21	G0470118	MASALA C	1	40	9.5	0	49.5	BEVERAGE
04-01-2010 13:21	G0470118	MOROCCA	1	45	10.69	0	55.69	BEVERAGE
04-01-2010 01:29	G0470284	CAPPUCCI	1	60	14.25	0	74.25	BEVERAGE
04-01-2010 01:29	G0470285	MOROCCA	1	45	10.69	0	55.69	BEVERAGE
04-01-2010 01:29	G0470285	SUMATRA	1	95	22.56	0	117.56	BEVERAGE

Figure 3.3: Cafe Dataset from Kaggle

- Above is one of the datasets used for this project which is acquired from Kaggle [ref no.] It is a cafe dataset and from the items mentioned, it was assumed that it is based in South India.

3.4.2 Data Pre Processing

We can't simply use the dataset as it is, so in order to use it efficiently the dataset undergoes a pre-processing stage.

The types of pre-processing the dataset has undergone are:

- **Standardisation:** Here the numerical attributes undergo Standardising, i.e, taking the mean of the entire column and finding the residual value between mean and each numerical attribute. The new value of the column is then calculated by dividing the residual value with the standard deviation of the whole column. This is done to ensure that all numerical value are under the same scale that is between the values -3 and +3 .

$$z = \frac{x_i - \mu}{\sigma}$$

where Z is the standardised value Xi is the actual value, μ is the mean value and σ is the standard deviation

- **One Hot Encoding:**It is a method to encode categorical data by converting each category of the columns into a separate column that contains the value 1 if that category exists for the given row and 0 if otherwise. In this project the method is taken a bit further by having values of the encoded categories of each of the columns represent quantities sold for that day.

- **Adding external attributes:** External attributes like temperature, if it is a weekend or not and similar data are merged with our dataset to increase the accuracy of prediction. A new column for 'Weekend' is created and filled with 0 or 1 depending on if that day is a weekend or not. The value is 0 if it is not a weekend and it is 1 if otherwise. Another column for 'Day' was added in. Here, the value of each block was updated with a number from 0 to 6 with respect to days from Sunday to Saturday.
- **Filling Missing Data:** Missing values such as average temperature and precipitation are filled into the dataset by taking the mean of the entire column.

3.4.3 Exploratory Data Analysis

This is where the pre processed data undergoes analysis to find patterns and trends.

Some of the analysis done during the course of the project are:

- **Analysed different items in different quantity ranges:** Data is analysed in such a way that the items sold in larger quantities are analysed together, the ones in moderate quantities are analysed separately and together and the same is done for the items with low selling rate. This is done to figure out which all items can be discarded (the ones sold in less quantities) so that the food wastage can be reduced and which all items should be produced at a larger rate (the high performing items) so that the success rate of the business is improved. ARIMA model was used for the implementation of the same.
- **Analysed the trend and seasonality using prophet model:** Prophet model takes two values: the date (dx) and the value to be predicted (dy). It uses the two values to learn and find the seasonality. It additionally assists the user in creating a visualization depicting the fluctuations in dy over both yearly and weekly periods, along with highlighting the trends identified within dy.
- **Analysed the normalised quantity of items sold in low and high temperature:** Once the columns for average temperature and precipitation were added, the quantities sold between each varying temperature and precipitation were found, visualized and analysed.
- **Finding the mean quantity sold on weekends:** ARIMA model was used to analyse the dataset and divide them into two groups: the one with items sold on weekdays and the one

with items sold on weekends. The data is analysed based on if it is a weekend or a weekday and then compared. It was found that the items were sold off at a larger and faster rate on weekends compared to weekdays.

- **Visualisation of quantity sold each month:** The dataset was categorized based on months. The data was then analysed month-wise. It was done based on items sold from April, 2010 to March, 2011. ARIMA model was used for the implementation of the same.

3.4.4 Model Identification

Here different models are tested out with the dataset acquired and the most accurate model for this dataset is found and used.

Some of models are:

- **Prophet Model:**
 - This Model takes input as dates and the output and gives a prediction output based on the input.
 - The prophet model is capable of generating a trend , yearly, weekly and daily change of item quantity which makes it quite useful for EDA.
 - It is a time series model which works well with datasets having strong seasonal effects and several seasons of historical data
- **Random Forest Model:**
 - It is an ensemble model, i.e a model that has more than one models that are trained together which in turn gives multiple output whose average is found as the predicted value.
 - Random forest contains multiple decision tree models for prediction
- **ARIMA model:**
 - Called as Autoregressive integrated moving average which uses time series data to better understand the data set or predict future data.
 - Mainly used for demand forecasting

- Potential Pros of ARIMA model are
 - * Only require prior data of a time series to forecast
 - * Performs well on short term forecast
 - * Models a non-stationary time series, i.e. the statistical properties of the data like mean and variance would not be constant over time
- Potential Cons of ARIMA model are
 - * Difficult to predict to predict turning points, i.e. the moment the direction of trend changes
 - * Computationally expensive
 - * Cannot be used for strong or too irregular seasonal time series data

- **Huber Regressor Model:**

- Huber Regressor is a linear regression model that is less sensitive to outliers compared to ordinary least squares regression. It achieves this by using a combination of squared and linear loss functions.
- It minimizes a squared loss with a squared regularization term for small errors and a linear loss for large errors. This combines the advantages of Ridge (L2 regularization) and Lasso (L1 regularization) regression models.

- **Bayesian Ridge Model:**

- Unlike ordinary linear regression, which uses a point estimate for the coefficients, Bayesian Ridge Regression uses a probabilistic approach. It estimates a probability distribution over the coefficients.
- It introduces regularization parameters to control the model complexity. It automatically adjusts the regularization strength based on the data.
- The Bayesian Ridge Regression model is robust to multicollinearity in the dataset. This means it can handle situations where features are highly correlated.
- The Bayesian Ridge Regression model is available in scikit-learn's linear model module. It can be used similarly to other regression models in scikit-learn.

- **KNN Regressor:**

- KNN Regressor is a non-parametric algorithm used for both classification and regression tasks. Unlike parametric methods like linear regression, KNN Regressor doesn't make any assumptions about the underlying data distribution.
- It predicts the target value for a data point by averaging the target values of its k nearest neighbors. The value of k is a hyperparameter that needs to be specified by the user.
- KNN Regressor is relatively simple to understand and implement. It relies on measuring the distance between data points in the feature space and selecting the k nearest neighbors based on this distance metric.

- **Gradient Boosting Machine:**

- Gradient Boosting Machines is an ensemble learning method used for both regression and classification tasks. It builds a strong predictive model by combining the predictions of multiple individual models, typically decision trees.
- GBM builds the ensemble sequentially, where each new model (learner) corrects errors made by the existing set of models. It uses a gradient descent algorithm to minimize the loss when adding new models to the ensemble. Popular implementations of GBM include XGBoost, LightGBM, and CatBoost, which offer efficient and scalable implementations of the algorithm.

- **AdaBoost Model:**

- AdaBoost is an ensemble learning method used for both classification and regression tasks. It works by combining multiple weak learners to create a strong learner. A weak learner is a model that performs slightly better than random guessing.
- AdaBoost trains a sequence of weak learners, where each new learner pays more attention to the instances that the previous ones misclassified. This is achieved by adjusting the weights of training instances after each iteration.
- In classification tasks, AdaBoost combines the predictions of weak learners using a weighted majority vote, where the weight of each learner's vote depends on its performance. In regression tasks, the predictions are combined by weighted averaging.

- **SARIMA Model:**

- SARIMA is an extension of the Autoregressive Integrated Moving Average (ARIMA) model that includes seasonality. ARIMA models are used to describe the autocorrelations in the data. SARIMA extends ARIMA by including additional parameters for seasonal components.
- SARIMA models take into account seasonal patterns in the data. These seasonal patterns are represented by three additional parameters: P, D and Q which are similar to the non-seasonal parameters p, d and q in the ARIMA model.
- Like ARIMA, SARIMA requires the input time series to be stationary. If the time series is not stationary, differencing (integration) is applied to make it stationary.
- SARIMA models are commonly used for time series forecasting, especially when the data exhibits seasonal patterns. They are available in Python through the statsmodels library.

Chapter 4

Results

A correlation analysis was conducted to assess the relationship between external factors and restaurant sales in the dataset. The analysis revealed both positive and negative correlations between various external factors and sales. For instance, a positive correlation was observed between factors like temperature, social events, and sales, indicating that as these factors increase, restaurant sales also tend to increase. Conversely, a negative correlation was found between factors such as precipitation and sales, suggesting that higher levels of precipitation are associated with lower restaurant sales. This correlation analysis provided valuable insights into how external factors impact restaurant sales, aiding in better understanding and forecasting sales patterns.

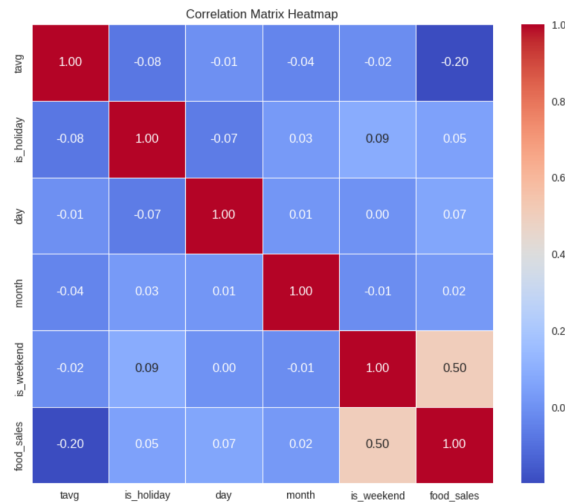


Figure 4.1: External Factors dependance on food sales

Various analysis graphs were generated using models such as ARIMA and Random Forest. These graphs helped visualize the performance of the models in predicting restaurant sales based on the provided data. The graphs provided insights into trends, patterns, and seasonality within the dataset, aiding stakeholders in understanding the effectiveness of different forecasting approaches. Additionally, these visualizations facilitated the comparison of model predictions with actual sales data, allowing for the assessment of each model's accuracy and reliability in capturing sales variability.

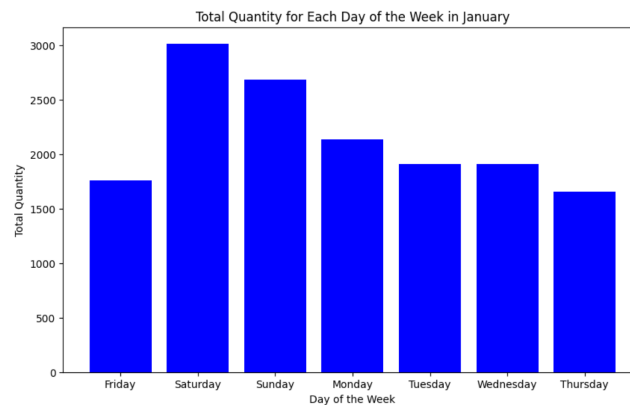


Figure 4.2: Weekly Sales in a month

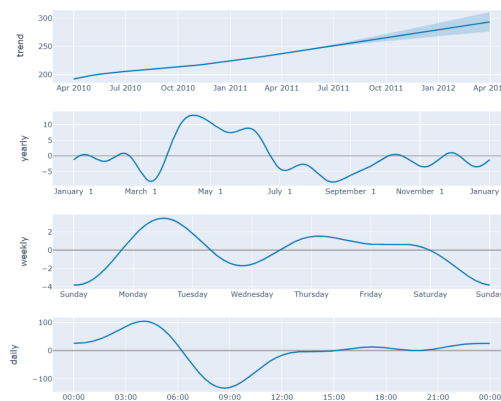


Figure 4.3: The graph depicting the trend,yearly,weekly and daily prediction of Quantity

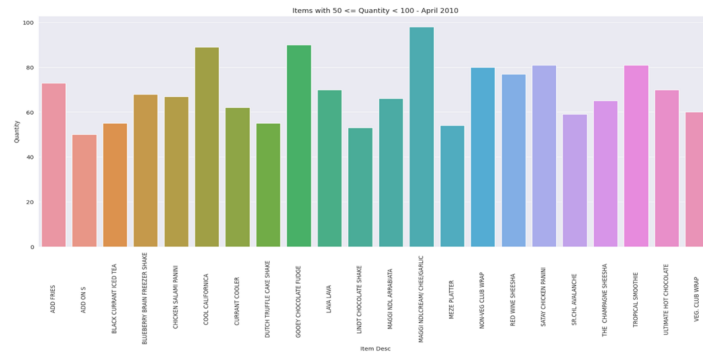


Figure 4.4: The graph depicting the items with quantities sold more than 50 and less than 100 in a month

A range of regression models, including Huber Regressor, K-Nearest Neighbors (KNN), Bayesian Ridge Regressor, Gradient Boosting, and AdaBoost, were utilized to calculate performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE) for predicting restaurant sales based on the provided data. These metrics were essential in evaluating the predictive accuracy of each model and identifying the most suitable approach for the task at hand.

Model	MAE	MSE	MAPE
Huber Regressor	0.6374	0.7560	1.0599
KNN	0.6605	0.7376	1.2370
Bayesian Ridge Regressor	0.6553	0.7457	1.1876
Gradient Boosting	0.6554	0.7332	1.2158
AdaBoost	0.6707	0.7467	1.2898

Table 4.1: ML models used

The dataset was processed using Seasonal Autoregressive Integrated Moving Average (SARIMA) before and after differencing. Interestingly, better results were obtained after differencing the dataset. Differencing helped stabilize the data by removing trends and seasonality, resulting in improved model performance. This indicates that differencing was effective in making the data more stationary, enabling the SARIMA model to better capture underlying patterns and make more accurate predictions.

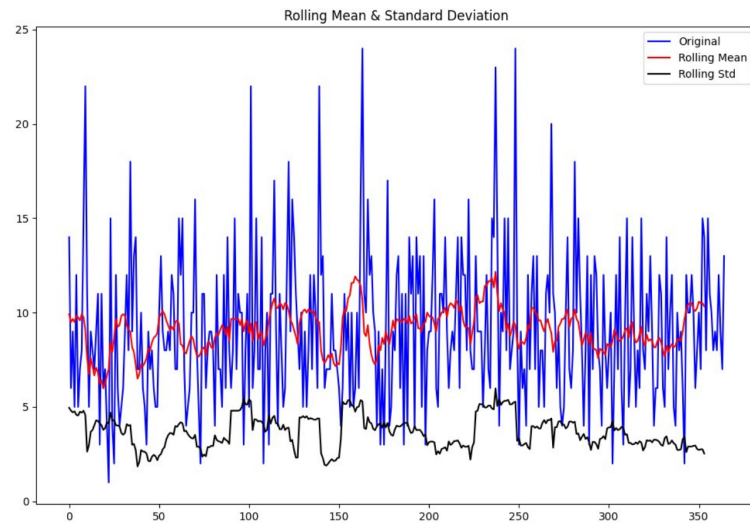


Figure 4.5: The graph before differencing was done

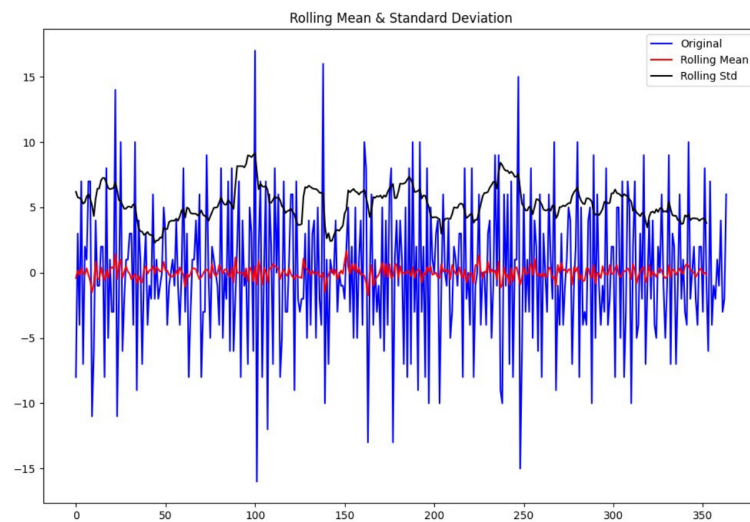


Figure 4.6: The graph after differencing was done

SARIMA (Seasonal Autoregressive Integrated Moving Average) was employed for forecasting restaurant sales data due to its ability to effectively capture the complex patterns and seasonality present in the dataset. SARIMA models are particularly advantageous for time series forecasting as they can account for both autoregressive (AR) and moving average (MA) components, as well as seasonality and trends through differencing. By considering these factors, SARIMA models provide more accurate and reliable forecasts compared to simpler models. Additionally, SARIMA

allows for the incorporation of external factors that may influence sales, providing a comprehensive forecasting approach that can adapt to the dynamic nature of restaurant sales data.

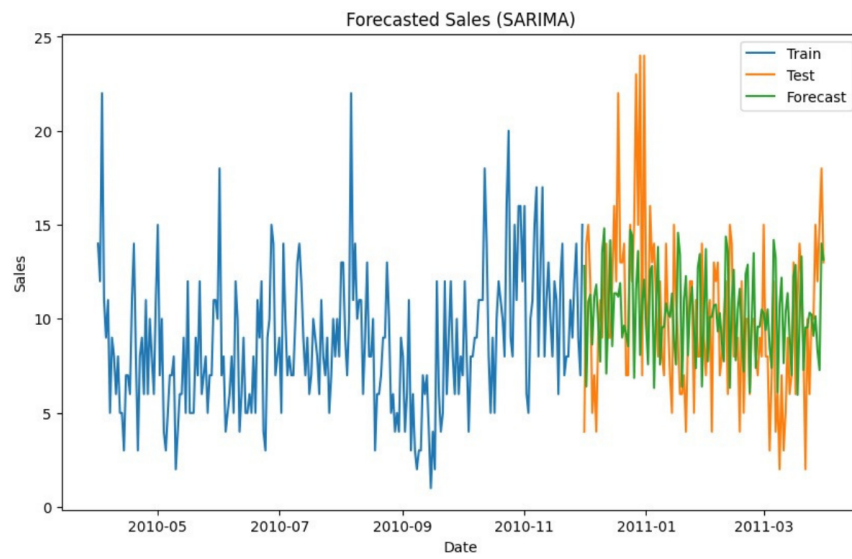


Figure 4.7: Forecasting using SARIMA

Chapter 5

Project Plan & Timeline

Project Phase	Date	Status
Dataset and Literature Survey	November	Completed
Model Identification	November	Completed
Model Training and Testing	December - January	Completed
Forecasting	February - May	Completed

Table 5.1: Project Plan and Timeline

Chapter 6

Task Identification & Allocation

Task	Team Member(s)	Expertise
Data acquisition	Abhay Siva, Ashly Ann Joseph, Junaid Sunil, Sreelakshmi Gireesh	Data collection and Preprocessing
Model Identification	Abhay Siva, Ashly Ann Joseph, Junaid Sunil, Sreelakshmi Gireesh	Model search, Evaluation, Selection
Model Training	Ashly Ann Joseph, Junaid Sunil, Sreelakshmi Gireesh	Code generation for training models
Model Testing	Abhay Siva, Ashly Ann Joseph, Junaid Sunil	Finding the accuracy of the model using test data
Forecasting	Abhay Siva, Ashly Ann Joseph, Junaid Sunil, Sreelakshmi Gireesh	Finding the accuracy of the model using test data
Documentation	Abhay Siva, Sreelakshmi Gireesh	Documenting the report and Creating presentations

Table 6.1: Task Identification and Allocation

Chapter 7

Conclusion

This project introduces a groundbreaking sales analysis system that leverages the power of Machine Learning (ML) and Artificial Intelligence (AI) to redefine how restaurants approach sales insights. Through a modular design, the system dynamically learns from diverse historical datasets, enhancing its predictive accuracy and allowing for customization based on the unique characteristics of each restaurant. The implementation of real-time analysis, interactive reporting, and robust data security measures ensures timely, user-friendly, and secure access to critical sales information. By prioritizing adaptability and self-learning mechanisms, this project not only addresses immediate analytical needs but positions itself as an evolving solution for the ever-changing restaurant industry landscape. This innovative methodology holds the potential to reshape how restaurants understand and optimize their sales strategies, fostering a more agile, informed, and successful business environment that aligns with the evolving needs of the industry.

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